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How Bird Songs Can Help Prevent Ecological Collapse

One of the most critical keystones of the ecosystem is in growing danger. Birds are being affected by global warming in unprecedented and unpredictable ways, and some scientists are turning to emerging technologies in automatic bird song recognition as a potential solution.

As the bitter chill of winter melts into the warmth of spring, birds all across North America begin their long migrations back home. While this astonishing phenomenon has been well documented for centuries, changes in the migratory patterns of bird species all across the country are beginning to concern and perplex scientists. As global temperatures continue to rise, researchers have found that birds have been migrating as much as two days earlier each spring, and to strange new locations [4]. This concerning discovery has led to a variety of novel methods to keep track of these migratory patterns. No longer do scientists rely on merely counting birds by eye and extrapolating their numbers as in previous decades. New technologies have allowed researchers to create complex maps and charts that can be used to analyze the behaviors of migratory birds and help us to understand the way in which their changing migratory patterns are affecting ecosystems all over North America. In the midst of this frenzied search for new methods of analysis, some researchers have turned toward an unexpected source: bird songs.

Migration and Ecological Stability

Not only is the time window of migration changing, its location is too. As climate change continues to cause global temperatures to climb each year, the area in which birds can migrate to is shrinking. Birds require a particular set of environmental conditions such as moderate temperatures, availability of mates, absence of predators, absence of adverse weather conditions such as storms, and ample food resources [1]. Climate change is affecting each of these in varying ways. Rising global temperatures have been pushing the area in which birds can migrate further toward the poles, both shrinking and moving the area suitable for breeding and

nesting further north and south [5]. By being pushed to migrate to areas that are both smaller and longer distances away, they are being forced to make more difficult and dangerous journeys to environments with higher amounts of predators, less food resources, worse weather conditions, and less mates. A study conducted in 2018 on European migratory birds found that over 80% of species will have significant increases in the distance and time it takes to reach their breeding ranges in the years to come [5]. This change can be seen in birds such as the thrush nightingale and the European bee-eater, with each predicted to face an extra 800-1000km of distance to their migration by 2070, coming out to an extra 5 and 4.5 days of travel respectively [5].

Timing is the most important factor for successful migration, as achieving a balance between arriving to breeding ranges when in the peak of reproductive health, and arriving early enough to secure a suitable breeding site, is critical. These necessities show that competition is the key factor of migratory success in birds [8]. The question of growing concern, then, is how shrinking and more distant breeding ranges will impact competition, and thus the reproductive success and survival of migratory birds.

The European Bee-Eater showing off its magnificent wingspan and vibrant colors

Pettersson. (2018, May). European Bee-eater [Photograph]. Retrieved from https://www.flickr.com/photos/130021529@N02/42975997664

The relationship between ecological stability and migration is reciprocal. This means that, as ecological instability threatens migratory behaviors, so too does their changing migratory behavior hasten ecological instability. This could lead to a devastating feedback loop with unseen consequences. Birds are among the most important organisms in maintaining the stability of an ecosystem. They are one of nature's great pollinators and fertilizers, carrying seeds, pollen, and vital soil nutrients across long distances on their journeys. They are so important that up to 92% of woody tree species have birds as their primary method of seed dispersion, carrying plants that are used for timber, medicine, food, and more, allowing them to spread along the length of their long migratory flight paths [6]. Birds are not only important for wildlife, but for humans as well. They are one of the most efficient forms of natural pest control that exist, with insect-eating birds maintaining the health of human crops throughout history. Birds are a common predator of invasive and devastating pests, such as crickets and beetles, and the stability of food supplies around the world is all due in part to the contribution of bird predation on these pests [6].

Automatic Song Detection

Perhaps one of the most novel methods of tracking migratory patterns for birds involves the use of bird song. Scientists have worked on developing systems that, when used in conjunction with specialized audio equipment called field recorders, may be able to successfully identify bird species based entirely off of their songs. Field recorders include a variety of robust audio recording equipment that are strategically placed in areas of interest in the wilderness. One such system, called "warbleR", utilizes field recordings in conjunction with computer processing and online databases to give estimates of how far the recorded species can be found geographically.

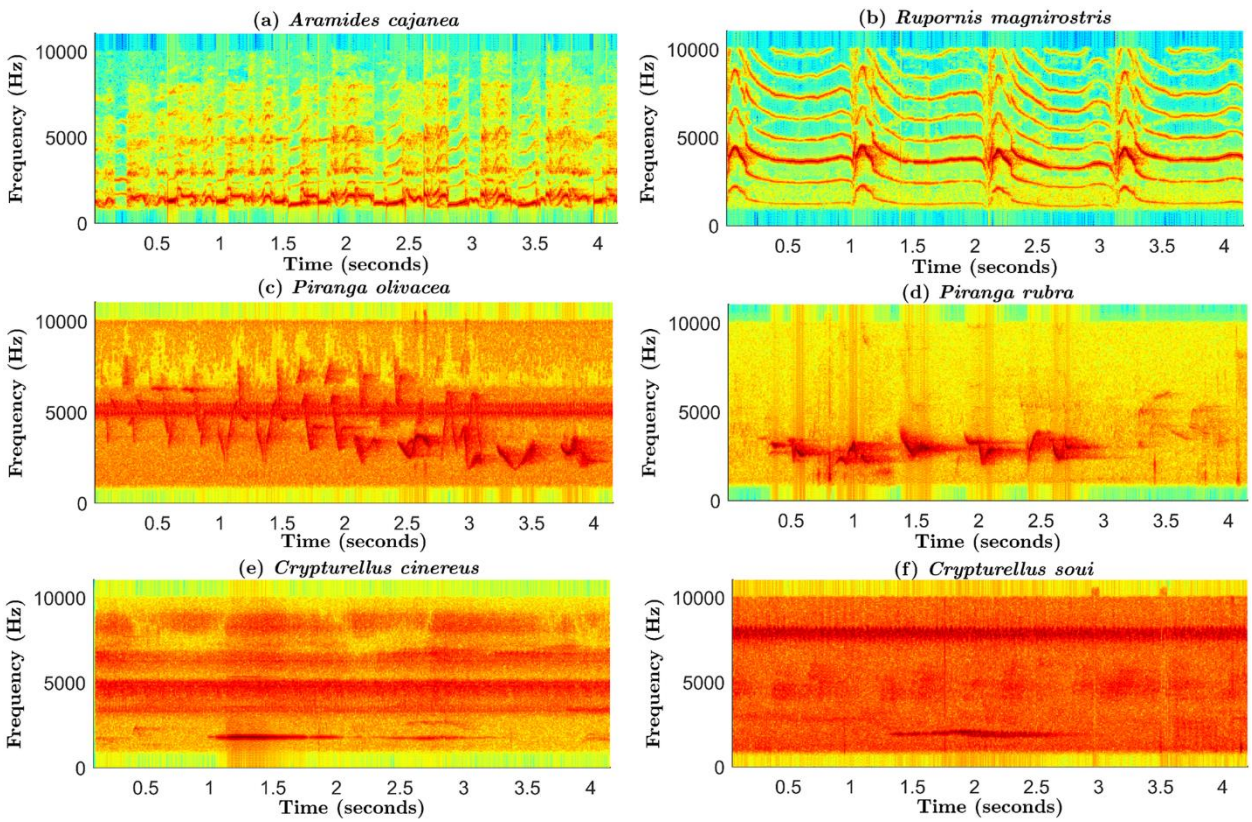
The unique characteristics of these recordings are used in a multistep process to identify species [2]. First, when the field recorders capture audio from their environment, they send the recordings to a program that then compares them to clean, already-identified recordings of the target species using an online database. These recorded songs are made up of syllables, like notes in a song, that have a particular sound characteristic that is unique to each species, which allows them to be identified apart from each other. The system takes clean

recordings from the online database and uses it to detect the exact syllables within the field recordings and identify the species [2]. Once this is completed, information related to geographic spread of the species can be compiled and graphically mapped out, showing researchers exactly where these birds can be found. These maps can then be used in conjunction with other environmental data to analyze migratory patterns.

Although not all of these systems are created specifically to create such maps, there is an entire world of software developed for analyzing bird song that can be readily fitted for studying migratory behavior. The field of bioacoustics, the study of sound in animals, has had a fixed interest in just such an application for over 20 years. Beginning in 1999 with software such as Sound Analysis Pro, the field of bioacoustics has blossomed into an ever-growing search for how to apply the latest cutting-edge developments in artificial intelligence, machine learning, and computer science, to work in automatically processing bird song [9].

Since its early beginnings, researchers interested in bioacoustics have utilized a wide variety of tools and methods to try to develop the quickest, most accurate, and most efficient system possible. The popularity of easy to use and collaborative programming languages have allowed for a boom in these systems within the past few years, allowing researchers all over the world to utilize and adapt new systems with ease [2]. To this day, hundreds of scientists around the world, driven by the increasing pressure of the disastrous effects of climate change, are hard at work developing the latest and greatest in bioacoustic technology.

The most recent developments have utilized artificial intelligence to look at visual graphs of sound over time called spectrograms, and visually compare their complex shapes in order to match them to known notes in bird songs and thus identify the bird [9]. This research has proven to be most promising, reporting higher and quicker success rates than any system yet developed [9]. With data gathered through such methods, researchers can construct accurate maps of where these birds are being identified at, and when. Compiled together this data can give researchers critical information to understand exactly how climate change is affecting these animals, and, hopefully, how to help stop it.



A typical spectrogram, comparing various bird species' songs

Ludeña-Choez, Quispe-Soncco, Gallardo-Antolín. (2017, June 19). Fig 7 [Illustration]. Retrieved from https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0179403

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